FROM MASS TO SURFACE

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MATERIAL, MEMORY, AND MODERNIZATION IN JAPAN AND KOREA

Abstract

A building, before anything else, is its material. Long before synthetic alternatives, earth was not chosen for a project; it was simply what a place had. Walls rose from the ground beneath them, thick enough to hold heat, keep the weather out, and register in their mass the weight and duration of the years they endured. Structure and site were inseparable. Material became a record of both place and time – of the labor, craft, and conditions through which a building came into being.

Historical architecture carries memory through these material traces. Its significance lies not only in form or design, but in the surfaces that weather, the techniques that endure, and the stories embedded in their construction. Earth once held these histories through mass: through walls whose thickness, texture, and imperfections recorded the process of making.

Modernization did not end that relationship so much as relocate it. As industrialized materials replaced earth's structural role, earth itself did not disappear. It moved toward the surface – to plaster, finish, and craft. Surface, as a material condition, can still carry the memory that earth once held as mass: technique stands in for thickness, refinement for weight. At the same time, contemporary technologies allow earth to become more than a historical remnant, opening new possibilities for its use in sustainable construction. To trace earth from mass to surface is therefore to ask how architecture preserves memory while adapting the materials through which that memory is expressed.



Tsubame-no-ochaya teahouse with skyscrapers in the background
Ginza, Tokyo

"Building with earth is an attitude. It means respecting cycles. It means not consuming materials, but using them—again and again. It means adapting to the site instead of dominating it. And it means taking responsibility – for what we leave behind."

- MARTIN RAUSCH

Upscaling Earth: Material, Process, Catalyst

Introduction

My mother remembers Korea in fragments: the countryside pressed up against the edge of the city, old construction still standing in the shadow of the new. She left when she was five and has never gone back. She still describes the layout of a house she has not stood inside in over four decades.

My father goes back to Japan as much as he can. Not to just the country, but to one city in particular: Tokyo. One place. He recalls his solo excursions back to his grandmother's home country. Growing up, he always told me about specific buildings he visited – how the rooms opened up in ways that stayed with him.

While my mother kept a place by leaving it and holding onto its memory, my father kept a place by returning to it, year after year, to the same spots. Between my parents, I learned that a home does not only live in the place itself, but it lives in what gets carried with us after the place is gone, or in what keeps drawing us back to it.

I grew up on the outside of both of these relationships, close enough to inherit the memories, but too distant to have formed my own. My sense of what a space means – what it holds and what it's for – was built from my mother's descriptions and my father's returns, long before I had the architectural language to depict and describe why any of it mattered.

The rupture between what was built and what came after it is not only a personal one. Japan and Korea were both pushed to rebuild quickly in the twentieth century, Japan after wartime destruction and Korea after the Korean War. Both moved away from earth as a structural material in the process. But the two countries diverged sharply in how that departure unfolded, much as my parents' two relationships to the home diverged, even as they were shaped by distance.

The Yale School of Architecture and the Harvey Geiger Fellowship allowed me to travel to Japan and Korea to trace that divergence directly, and in a sense, to encounter for myself the places I had first inherited through someone else's memory. This helped me address my own question: how did Japan and South Korea's differing paths toward modernization shape their relationship to earth as a building material?



ACKNOWLEDGEMENTS

To Mr. Geiger

Thank you Mr. Geiger for supporting my travel and research this summer. Without your generosity, I wouldn't have been able to travel across Japan and Korea.

There is a kind of understanding that only comes from physical presence that no photograph or book could have given me. This experience instilled in me that architecture cannot be fully understood at a distance but through firsthand observation. I hope to pass this insistence on to future generations of Yale architecture students, the way you have for mine.

To Dad

Thank you for always pushing me to delve deeper into my curiosities, especially my fascination with architectural design, and for giving me the travel bug. Your love for Japan – and for the world – inspired this trip, and I am thankful I got to share this story with you. Thank you for pushing me not just to explore beyond my limits, but to grow through the act of traveling itself, and for always encouraging me to pursue my dream of studying architecture.

To Mom

Thank you for sharing your stories about your childhood in Korea, even the ones softened and reshaped by distance and time. Hearing you describe a home you left and never truly returned to inspired me to form my own connection to my Korean culture. I am endlessly grateful for the home you gave me, built in part from the one you had to leave behind.

Extending my thank you's

To Surry and Professor Dugdale, for encouraging me to be curious in and out of the classroom, and for seeing the potential in this project before I fully did.

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To Koichi Yamagishi, my Japanese tour guide and Mr. Donut connoisseur, for letting me crash on your couch while I traveled across Japan. Thank you for all the delicious meals, hidden sushi spots, and for showing me a side of Japan I would have never found on my own.

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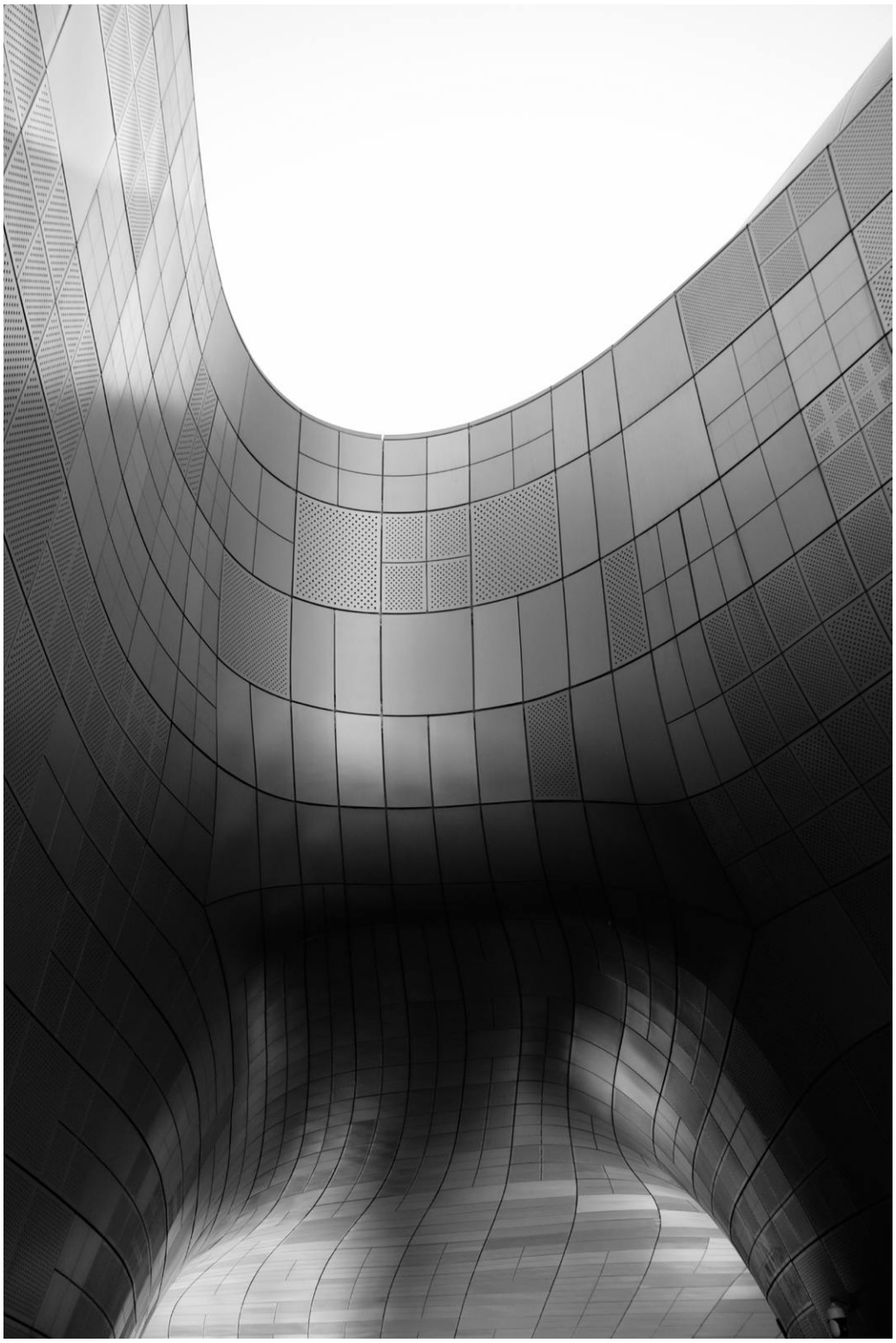
This research trip carried forward the questions posed in my initial proposal: How did Japan and South Korea diverge in terms of their modernization timelines? And how did that divergence shape their relationship to earth as a building material? I wanted to trace what shifted — earth as structure, earth as surface, or earth's absence altogether — and what that shift reveals about each country's negotiation between tradition and modernity.

Over six weeks and eight regions across both countries, I sought out sites where modern materiality and earth-based construction sit in direct proximity — often at the edges of major cities, where the pressure of development is felt but has not yet erased vernacular practice.

POST-WAR RECONSTRUCTION AS A SHARED CATALYST

A theme that became clearer through travel than it had been on paper was the shared role of rapid reconstruction. Japan and Korea each faced urgency to rebuild — Japan after wartime destruction, Korea after the Korean War — and in both cases, that urgency favored fast, standardized construction over the slower, labor-intensive processes tied to earthen building.

This pressure helps explain why both countries departed from traditional material practice in their urban centers during the twentieth century, even as the paths that departure took diverged sharply. Nowhere was this divergence clearer than in the contrast between my first days in each country. In Tokyo, the departure from earth reads as refinement — materials didn't disappear so much as retreat into ritual and craft. In Seoul, my first encounter was Zaha Hadid's Dongdaemun Design Plaza, its neofuturistic aluminum and steel skin curving without a single straight line — a building that doesn't relocate earth's logic at all, but replaces it outright with an entirely synthetic, computationally-driven language. Standing in front of it, I understood the two countries' shared catalyst had produced two different endpoints: Japan relocating earth to the surface, Korea in places abandoning the material register of earth altogether in favor of the synthetic and the futuristic.



Dongdaemun Design Plaza by Zaha Hadid
Seoul, South Korea





Tofukuji Temple's traditional wooden bracket roof
Kyoto, Japan

JAPAN: FROM STRUCTURAL EARTH TO SURFACE, CRAFT, AND CONCRETE

Within Tokyo, structural earth is nearly absent, replaced by engineered timber, steel, and concrete systems shaped by seismic regulation and early industrialization. But even inside this density, earth persists, not as structure but as a deliberately preserved register of craft.



At Hamarikyu Gardens, walking past the Nakajima-no-ochaya teahouse, I found the clearest embodiment of my hypothesis. The teahouse, first built in 1707 and rebuilt in 1983 after fire, was reconstructed using its original dimensions, materials, and technique: colored earthen-wall plaster made from Osaka clay, hari-tsuke kabe paper pasted onto boards and fixed with lacquered timber, a ceiling of thousand-year-old yakusugi cedar. Nothing here is structural in the way a hanok wall is structural — the teahouse survives inside a park, itself an island inside the city, the way the proposal predicted earth would survive in Japan: as ritual, not necessity.







I saw the same logic again at the Kyu Asakura House in Daikanyama, where craftsmanship I'd only read about at Hamarikyu's teahouse was suddenly legible at the scale of joints and finishes.







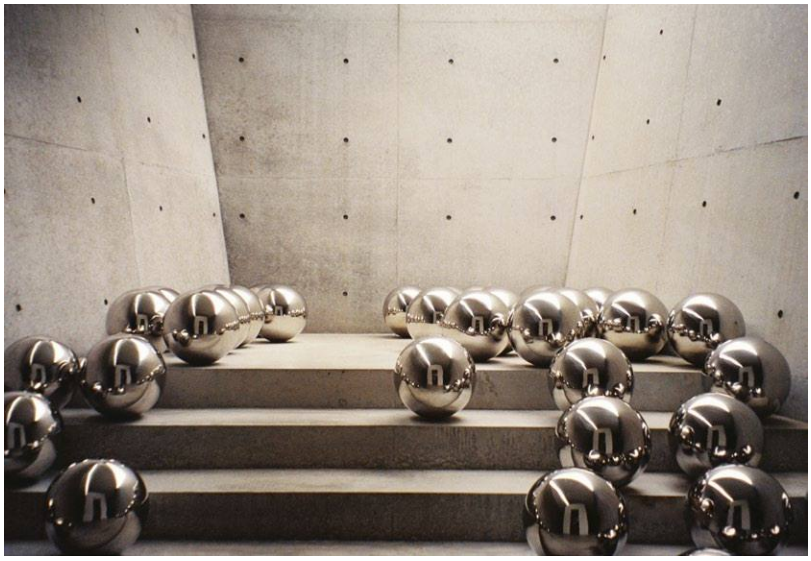
Naoshima offered the sharpest test of my central claim about concrete as "abstracted, smoothed earth." Tadao Ando's architecture on the island reads as a direct descendant of rammed earth's mass and interiority, its sedimentation and irregularity replaced by a controlled, minimal surface — but the relationship to weight and enclosure remains.



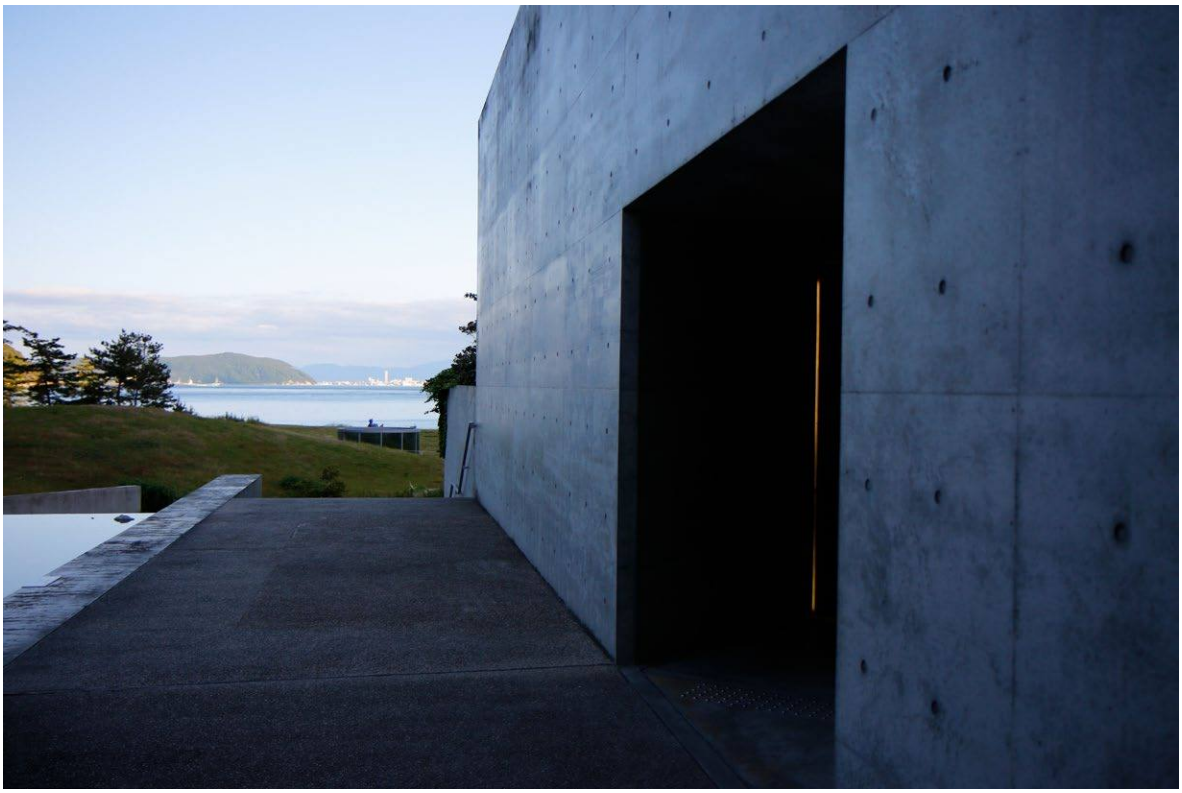
What surprised me on Naoshima was how much of the island's material story is not earth or concrete at all, but synthetic material doing the emotional and structural work earth once did.



SANAA's bubble-shaped bicycle shelter is fiber-reinforced plastic stretched over a wooden grid.



Even Kusama's Narcissus Garden, a field of mirrored stainless steel spheres, seemed to ask what it means to occupy natural landscape with something entirely fabricated, multiplying the artist's and the viewer's own reflection back at themselves.



Ando's Benesse House Museum overlooking the sea
Naoshima, Japan





Stark material and vernacular contrast to Naoshima's residential neighborhood.







Post-WWII reconstruction in Japan was shaped as much by policy as by necessity. The government's policy standardized construction nationwide, a shift that effectively erased vernacular timber-framed building techniques and systems that had been practiced regionally before the war.

Structural earth and traditional timber framing weren't abandoned by accident; they were legislated and engineered out of the structural system, which is precisely what makes their survival in spaces like the teahouse and shrines — protected, ritualized, deliberately exempt from that system — so pointed.

Kyoto, however, offers a different kind of survival. Removed from America's bombing targets for its cultural and historical significance, the city was largely spared the destruction that reshaped Tokyo, leaving much of its traditional shrine and temple framing not rebuilt in exemption from modern code, but simply untouched by the war that necessitated it.



Ryoan-ji Temple
Kyoto, Japan





Tsutenkyo Bridge
Kyoto, Japan

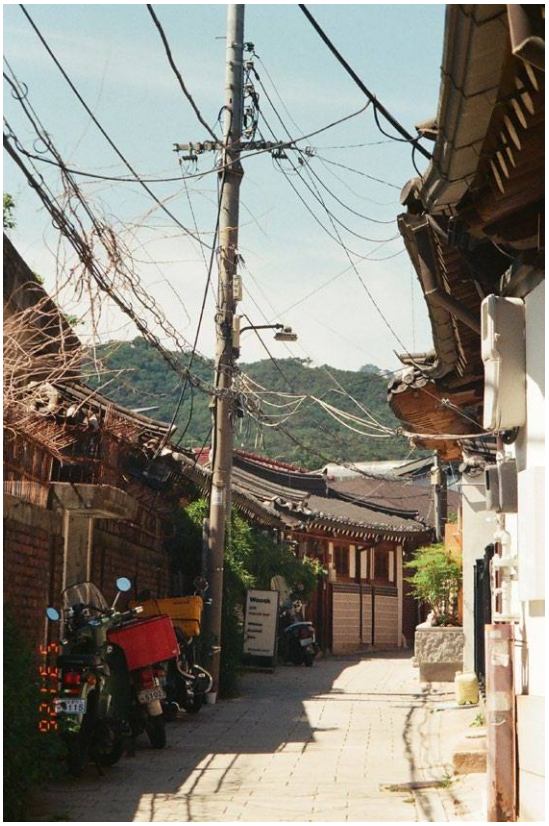
KOREA: THE PERSISTENCE OF EARTH AS MASS



Wongaksa Ten-Story Pagoda sealed inside glass tower to protect the monument from pollution. Preserving the original with synthetic.

Korea's departure from earth followed a different mechanism entirely — one driven less by building code than by state ambition. Following the Korean War, mass housing became a tool for projecting the image of a modern state; concrete apartment living was positioned as the image of progress itself, not simply a response to a housing shortage. Traditional hanok neighborhoods survive today only where they were deliberately preserved, rather than where they were structurally permitted to remain.


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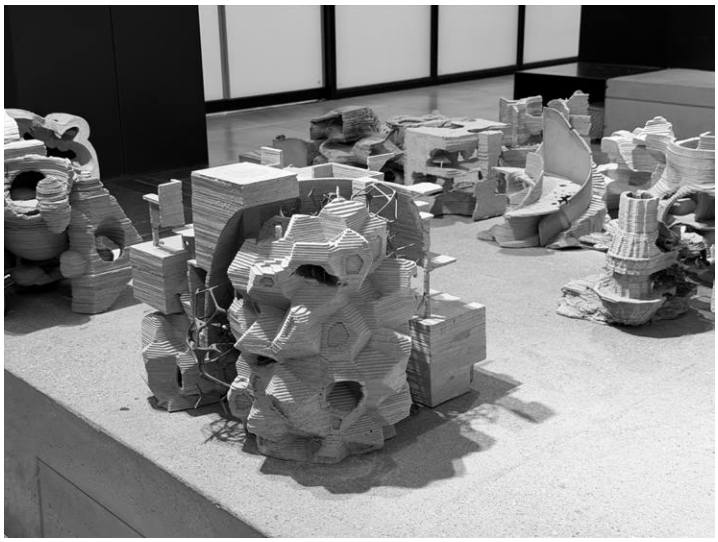

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Seoul's everyday city life is layered and improvised with storefronts, signage, and air conditioning units accumulating onto building facades as functional necessity, with little attention paid to surface as anything beyond utility. In contrast, Gyeongbokgung Palace treats ornamentation as its own structural language: dancheong's painted brackets and tiered eaves are maintained with the same precision as the building's load-bearing elements, so that decoration itself becomes a form of historical memory rather than an afterthought.

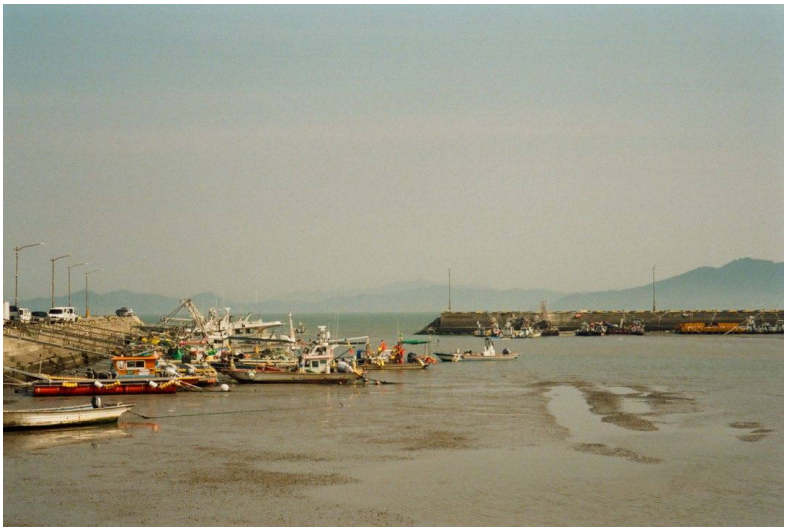
At Hongik University, that same tension appeared from the opposite direction. Undergraduates were 3D-printing concrete into organically shaped structures — learning to design from computational models rather than from the ground itself. Where Wongaksa protects the old material by enclosing it, the university is generating an entirely new material language, one that borrows the appearance of organic form without any of earth's original constraints.







Traveling outside the capital made earth's persistence far more visible than Seoul alone could show. In Boseong, land has been cultivated specifically into tea fields since reconstruction after the Korean War — climate and terrain shaping the landscape as directly as any wall.



In Jeju, an entirely different relationship to material emerged: the island's economy and imagery are built from the sea and its resources, almost as if the fish become the material.



Craft, material, and landscape remain in close conversation — echoed once more in the Bonte Museum, where Ando's concrete reappears, facing the island's coastline the same way it faced on Naoshima.





Before this trip, I understood earth's shift from mass to surface as a historical fact I was documenting from the outside. Traveling through it, I started to understand it instead as a design decision I am now responsible for making myself, every time I choose a material.



Go'O Shrine
Naoshima, Japan



The distinction — between a material's origin and the discipline it's asked to perform — is the one idea from this trip I expect to shape every project I take on from here forward. I no longer think of "sustainable" and "traditional" as separate design vocabularies. A material earns its authenticity through its relationship to site and to time, not through whether it came from the ground.

I arrived expecting to find nostalgia for earth wherever craft persisted, and technological detachment wherever synthetic material took over. Instead, I found something closer to inheritance under different conditions — the same impulse to make land, memory, and material speak to each other, carried forward through whatever tools a generation has.



The story of modernization in Japan and Korea is not a story of loss, but of relocation — earth doesn't disappear so much as migrate into new registers, from mass into craft, from structure into surface, from literal material into synthetic material asked to behave like it.



I too want to treat material choice as an active argument about memory and site, not a neutral technical decision. This trip gave me the vocabulary — mass, surface, relocation, inheritance — to think and design that way going forward, and the photographs gathered across these six weeks are the beginning of testing that vocabulary at the scale of an actual building.